



AutoTech Manufacturing Corp.

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IATF 16949 Certified | Engineering Division

BILL OF MATERIALS (BOM)

DOCUMENT NO.
AT-BOM-2401

REVISION
Rev 1.2

BOM INFORMATION

BOM Revision:	Rev 1.2
BOM Date:	October 10, 2025
Status:	Approved
Effective Date:	October 10, 2025
Supersedes:	Rev 1.1 (Sep 24, 2025)
Document No.:	AT-BOM-2401

REFERENCE INFORMATION

Customer PO:	PO-2025-15903
Customer:	AutoTech Manufacturing Corp.
Project Code:	PRJ-AT-2025-Q4B
Prepared By:	Technical Dept.
PO Line Item:	Line 1 - 200 pcs
Supplier:	Precision Steel & Components Ltd. (SUP-PSC-2024-456)

MATERIAL IDENTIFICATION

ID	TYPE/PART DESIGNATION	DIMENSIONS	MATERIAL GRADE
MAT-2401	Aluminum Angle - L Profile	50 x 50 x 5 mm x 6000 mm	6061-T6

Material Overview: Extruded aluminum equal angle with L-shaped profile. Suitable for structural applications, framing, and support systems. Material conforms to ASTM B221 specifications.

TECHNICAL SPECIFICATIONS

Material Standard:	ASTM B221 - Aluminum Alloy 6061-T6	Alloy Grade:	6061-T6 (heat-treated and artificially aged)
Leg Length:	50 mm x 50 mm (equal angle)	Wall Thickness:	5 mm
Length:	6000 mm (6 meters)	Surface Finish:	Mill finish (as-extruded)
Thickness Tolerance:	±0.3 mm	Length Tolerance:	+10 mm / -0 mm
Weight per Meter:	1.33 kg/m (approx.)	Testing Required:	Material certificate, dimensional inspection
Profile Type:	Equal angle, sharp outside corner	Leg Tolerance:	Per ANSI H35.2, standard
Straightness:	Max. 1.0 mm per 300 mm length	Twist:	Max. 1° per 1000 mm length
End Condition:	Saw cut, deburred	Temper Verification:	Hardness / conductivity check

CHEMICAL COMPOSITION (SPEC. LIMITS)

Element	Requirement (wt %)
Silicon (Si)	0.40 - 0.80
Iron (Fe)	0.70 max
Copper (Cu)	0.15 - 0.40
Manganese (Mn)	0.15 max
Magnesium (Mg)	0.80 - 1.20
Chromium (Cr)	0.04 - 0.35
Zinc (Zn)	0.25 max
Titanium (Ti)	0.15 max
Aluminum (Al)	Remainder

MECHANICAL PROPERTIES (MIN.)

Property	Requirement
Tensile Strength	260 MPa (38 ksi)
Yield Strength (0.2%)	240 MPa (35 ksi)
Elongation in 50 mm	8 %
Hardness (typical)	95 HB
Density (typical)	2.70 g/cm ³

Minimum values per ASTM B221 for 6061-T6 extruded profiles, wall thickness 1.60 - 6.30 mm. Typical values for reference only.

QUALITY & INSPECTION PLAN

#	Characteristic	Method / Reference	Frequency	Record
1	Chemical composition	Mill certificate review vs. ASTM B221	Each heat/lot	Material cert.
2	Mechanical properties	Tensile test per ASTM B557	Each lot	Material cert.
3	Leg length & thickness	Caliper / micrometer	5 pcs per bundle	Inspection report
4	Length	Tape measure, calibrated	5 pcs per bundle	Inspection report
5	Straightness / twist	Surface plate & feeler gauge	2 pcs per bundle	Inspection report
6	Surface condition	Visual, no dents, scratches, corrosion	100 %	Packing list sign-off



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PACKAGING, MARKING & HANDLING

Bundling:	Bundles of 20 pcs, strapped with non-metallic banding at 1.5 m intervals
Protection:	Interleaved with paper; ends capped; wrapped in VCI poly film
Marking:	Ink-jet marked every 1 m: alloy/temper, heat/lot no., ASTM B221, mill ID
Bundle Tag:	Material ID MAT-2401, PO-2025-15903, qty, lot no., gross weight
Storage / Handling:	Store indoors, dry; lift with nylon slings only - no chains or hooks

ADDITIONAL SERVICES & FEES

Fee (per supplier price list)	Applies	Charge basis	Quantity	Notes
Cutting Fee	Yes	Per cut	200 cuts	1 cut per bar: trimmed square to exact 6000 mm (+10 / -0 mm) and deburred
Cert Fee (MTC 3.1)	Yes	Per heat/lot	1 certificate	All 200 pcs supplied from a single heat/lot
Testing Fee	No	-	-	Dimensional inspection is part of standard supplier QC; not charged
Hydrostatic Testing	No	-	-	Not applicable to this material
Packaging (Standard / Export Grade)	No	-	-	Standard strapped bundles; no pallet or crate charge
Rush / Same-Day Processing	No	-	-	Standard lead time; no surcharge

Fee names and rates as listed in the Precision Steel & Components Ltd. Master Price List, effective October 01, 2025. Fees marked **Yes** are charged in the quantity shown and are **not** included in the PO unit price or PO total; they are invoiced as separate lines. Fees marked **No** are not charged for this item. No volume discount applies.

REVISION HISTORY

Rev	Date	Description of Change	By	Appr.
Rev 1.0	Sep 08, 2025	Initial release for quotation Q-PSC-25-3310.	D. Kowalski	R. Mendez
Rev 1.1	Sep 24, 2025	Length tolerance revised to +10 / -0 mm per customer request.	D. Kowalski	R. Mendez
Rev 1.2	Oct 10, 2025	Linked to PO-2025-15903; added packaging and inspection requirements.	D. Kowalski	R. Mendez

CONTACTS & APPROVALS

Role	Name / Department	Signature	Date
Prepared	Technical Dept. - Daniel Kowalski	/s/ Daniel Kowalski	Oct 09, 2025
Checked	Laura Chen, Quality Engineer	/s/ Laura Chen	Oct 10, 2025
Approved By	Engineering Department - Robert Mendez, Chief Engineer	/s/ Robert Mendez	Oct 10, 2025
Technical Contact	Thomas Richardson, Technical Sales Precision Steel & Components Ltd. (Supplier)	t.richardson@precisionsteel.com	+1 (412) 555-8912

Document Control: This BOM is controlled by AutoTech Manufacturing Corp. Material must meet ASTM B221 6061-T6 specifications.